Points to Remember

- Nervous system controls and coordinates the activities of our body.
- Neuron is the structural and functional unit of the cell and has three parts- cyton, dendrites and axon.
- A receptor is a cell or group of cells that receives the stimuli. An effector is a part of the body which can respond to a stimulus according to the instructions from the brain or the spinal cord.
- CNS is formed of brain and spinal cord. PNS consists of all nerves which connect brain and spinal cord to all parts of the body. ANS operates automatically and formed of sympathetic and parasympathetic nerves
- A reflex action is a rapid, automatic response to a stimulus which is not under the voluntary control of the brain.

TEXTBOOK EVALUATION

I. Choose the correct answer

1. Bipolar neurons are found in
   (a) retina of eye (b) cerebral cortex
   (c) embryo (d) respiratory epithelium
2. Site for processing of vision, hearing, memory, speech, intelligence and thought is
   (a) kidney (b) ear
   (c) brain (d) lungs
3. In reflex action, the reflex arc is formed by
   (a) brain, spinal cord, muscle
   (b) receptor, muscle, spinal cord
   (c) muscle, receptor, brain
   (d) receptor, spinal cord, muscle
4. Dendrites transmit impulse _____ cell body and axon transmit impulse _____ cell body.
   (a) away from, away from
   (b) towards, away from
   (c) towards,towards
   (d) away from, towards
5. The outer most of the three cranial meninges is
   (a) arachnoid membrane (b) piamater
   (c) duramater (d) myelin sheath
6. There are _____ pairs of cranial nerves and _____ pairs of spinal nerves.
   (a) 12, 31 (b) 31, 12 (c) 12, 13 (d) 12, 21
7. The neurons which carries impulse from the central nervous system to the muscle fibre.
   (a) afferent neurons (b) association neuron
   (c) efferent neuron (d) unipolar neuron
8. Which nervous band connects the two cerebral hemispheres of brain?
   (a) thalamus (b) hypothalamus
   (c) corpus callosum (d) pons
9. Node of Ranvier is found in
   (a) muscles (b) axons
   (c) dendrites (d) cyton
10. Vomiting centre is located in
    (a) medulla oblongata (b) stomach
    (c) cerebrum (d) hypothalamus
11. Nerve cells do not possess
    (a) neurilemma (b) sarcolemma
    (c) axon (d) dendrites
12. A person who met with an accident lost control of body temperature, water balance, and hunger. Which of the following part of brain is supposed to be damaged?
    (a) Medulla oblongata (b) cerebrum
    (c) pons (d) hypothalamus

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